

Sinus Node Dysfunction in Competitive Athletes

A CASE REPORT AND CURRENT RECOMMENDATIONS

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ABSTRACT

Cardiovascular screening of a player is essential to prevent sudden cardiac arrest in athletes and to an extent, electrocardiogram (ECG) screening enables to distinguish pathological changes from physiological changes related to training. Identifying the athletes at high risk of cardiac disease is critical due to paucity of evidence. The present study describes a case of asymptomatic sinus node dysfunction in an elite Volleyball player with suspected ECG findings during cardiovascular screening. The current recommendations and future research regarding their management and sports participation issues are also discussed.

Key words

Bradycardia, Sinus pause, ECG screening, Holter monitoring

INTRODUCTION

Physical training, especially aerobic training, induces functional and morphological changes in the heart that can be reflected in the electrocardiogram (ECG) (Magalshi et al, 2011). It is generally accepted that individuals who aspire to competitive sports should undergo pre-participation cardiovascular assessment (Corrado et al, 2005). Nonetheless, the optimum screening strategy for prospective athletes remains unresolved (Corrado et al, 2005; Estes et al, 2001). Further, as pointed out in the 36th Bethesda Conference, the published recommendations that do exist for identifying individuals at 'high risk' for adverse sport-related cardiac health issues are based mainly on expert consensus of 'what seems reasonable' (Zipes et al, 2005).

Rhythm disorders and changes in the

repolarisation pattern identified by electrocardiogram (ECG) screening are the most common ECG findings in sports person. Prevalence of rhythm disturbances including moderate sinus bradycardia (Heart Rate 40–50 BPM) is high (50–100%), especially in highly trained aerobic resistance sports (Cycling, Marathon, and Skiing). Even Sinus arrhythmias are observed in 15–20% elite sportspersons and so are sinus pauses. A sinus pause or arrest is defined as the transient absence of sinus P waves on the ECG that may last from two seconds to several minutes.

In as much as there is concern that physical exertion may trigger potentially hazardous tachyarrhythmias, pre-participation screening has principally focused on tachycardia susceptibility. Bradyarrhythmias have received less

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attention in as much as they are presumed to be less likely to pose a risk during physical activity. Certain bradyarrhythmias (first degree and Mobitz type I atrioventricular (AV) block) are common in the resting competitive athlete and are generally deemed to be of little concern in the absence of underlying structural heart disease (Ector et al, 1984). Sinus Node Dysfunction (SND) refers to a wide range of abnormalities involving sinus node and atrial impulse generation/ propagation (Dobrzynshi et al, 1932; Epstein et al, 2008). The pathophysiologic mechanisms for SND include failure of impulse generation and failure of the impulse to spread throughout the atria⁴. Symptoms are diverse and range from nonspecific complaints of fatigue, palpitations, irritability, lassitude, lack of concentration, forgetfulness to the more dramatic symptoms of syncope, recurrent dizziness, and heart failure (Alboni et al, 2009; Peters et al, 2008). Although SND occurs at any age, the incidence increases exponentially with advanced age (Dobrzynski et al, 2007), the mean age for diagnosis is 68 years, and both genders are equally affected (Adan & Crown, 2003; Viayaraman & Ellenbogen, 2011).

In this case report, we describe the management of an elite Indian player with suspected sinus pause and appropriate testing to evaluate the cause and extent of the underlying condition.

CASE REPORT

A 27 year old Indian male Volleyball player, training since 12 years, was selected for Pre-Asian Games national camp at NSNIS, Patiala. He underwent cardio

vascular screening including 12 lead ECG, as a part of pre-participation medical assessment in the Department of Sports Medicine. His history and clinical examination revealed no significant symptoms and signs related to cardiac disease. He reported no family history of heart disease or cardiac symptoms. A resting 12 lead ECG was done which demonstrated transient absence of P waves in lead II, as shown in figure 1, suspicious of sinus pause. An echocardiogram was considered because it can document the presence of underlying valvular or ischemic heart disease and may suggest the diagnosis of amyloid when diffuse conduction system findings are present. His echo revealed no structural abnormality. An ambulatory 24 hour ECG monitoring [Holter] was also advised to monitor timing and duration of pause, average heart rate ectopics and to establish arrhythmia symptom correlation if any. Holter ECG monitoring reported an average heart rate of 49. The minimum heart rate was 28 at 15:23 min, as shown in figure 2 and 3. The maximum heart rate was 104 recorded at 00:35 min. Pauses of different duration were recorded but none of them were greater than 2.5 seconds. Ventricular ectopy was 19, and supraventricular ectopy was 169, with 8 SV-runs, No ventricular and supra ventricular bigeminy and trigeminy events reported. No fibrillation and flutter reported. ST episode minutes totalled 2. The maximum absolute ST change was -2.7 and occurred at 03:19. He also demonstrated a good heart rate response to exercise stress. He was hence cleared for sports participation and advised regular followups.

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Fig.- 1: ECG demonstrating absent P wave

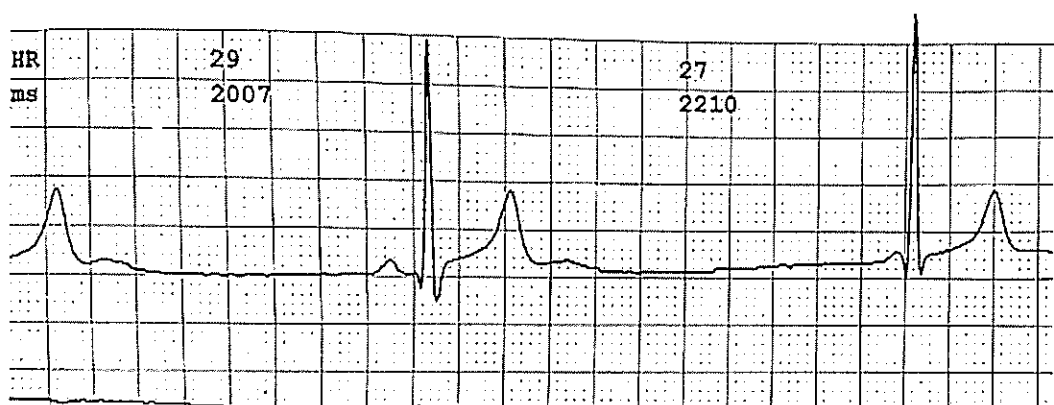


Fig.- 2 : Bradycardia(RR interval) observed in Holter recording

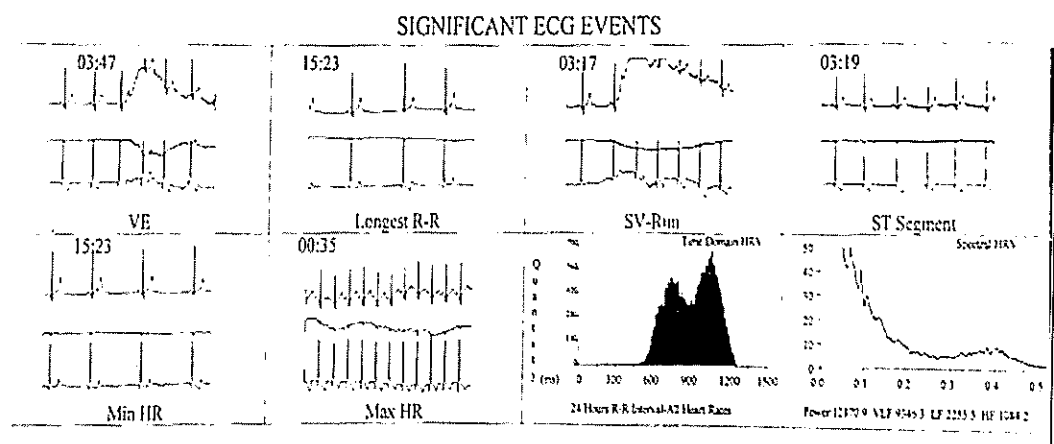


Fig.- 3: Significant ECG events observed in Holter recording

DISCUSSION

The factors accounting for cardiac pauses in a given player warrant individual evaluation prior to considering potential therapeutic options. In this regard, enhanced resting parasympathetic tone (or perhaps more accurately enhanced parasympathetic vs. sympathetic balance) is an accepted adaptation to repetitive exercise in competitive athletes (Tunay et al, 2016.) Prolonged sinus pauses, especially during rest or sleep, may be an expected rhythm variant in well-trained individuals. However, high-grade AV block during exercise, chronotropic incompetence, or strain-induced pauses (e.g. during weight lifting) would reasonably be considered to be abnormal, and may signal underlying disease.

Distinguishing the abnormal cardiac pauses which merit treatment, from more benign forms is not straightforward. Even discovering the presence of underlying structural heart disease cannot definitively define the risk associated with an observed pause. It is evident from a recent systemic review that duration of the pause (>3 s) is an inadequate marker. Moreover, competitive athletes may present with pauses > 3 s at rest. Presence of a "threshold pause" should be established in studies with extended follow-

up, since 3 seconds do not seem to discriminate players with or without future abnormalities.

There is not enough evidence or randomized controlled studies demonstrating the medical clearance, initiation of management and indications of pacemaker in players with pause. Transient interruption of training could, in many cases, banish longer pauses or decrease them in number and or time between RR intervals. It does not seem to be the most reasonable approach without structural heart disease and with adequate heart rate response to exercise, where no secondary effects have been demonstrated, due to the presence of prolonged sinus pauses.

CONCLUSION

As the published evidence based on sinus pause in sportspersons is limited, future reports should focus on follow-up of such patients, over many years, in order to more clearly establish outcomes. Till further studies clarify indications for initiating treatment for detected bradycardia and pauses, the most important diagnostic step remains the establishment of arrhythmia-symptom correlation, during ambulatory cardiac monitoring.

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